MARKET REQUIREMENTS DOCUMENT (MRD)

AquaGuard AI: Market Requirements Document

Fish Disease Risk Detection & Early Warning System • Commercial & Economic Architecture

PROJECT
AquaGuard AI – Smart Aquaculture Early Warning

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Target Market: Global Inland Freshwater Aquaculture (Focus: South & Southeast Asia, Sub-Saharan Africa)

1. Executive Summary & Market Opportunity

1.1 The Macroeconomic Imperative

Aquaculture is the fastest-growing agri-food sector in the world, now providing **over 50% of all aquatic food consumed globally** (FAO SOFIA 2024 Report). By 2030, global aquatic food production must expand by an additional 15% to meet the dietary protein requirements of a growing global population under the United Nations Sustainable Development Goals (SDG 2: Zero Hunger; SDG 14: Life Below Water).

However, biological instability and infectious disease outbreaks represent the single greatest operational and financial threat to this expansion. According to assessments by the **World Bank and the Food and Agriculture Organization (FAO)**: * **Global Annual Direct Losses:** Over **$6.0 Billion USD** in direct fish and crustacean biomass mortality per annum. * **National Sector Tolls:** In India, aquatic animal diseases inflict an estimated annual loss of **$2.48 Billion USD** (~15% of total sector output value); in Vietnam, annual losses exceed **$1.0 Billion USD**. * **Microeconomic Impact:** For a typical smallholder operating a 1-hectare earthen pond, a single uncontrolled disease outbreak (e.g., *Aeromonas* septicemia, White Spot, or Columnaris) results in an 80–100% crop loss, wiping out an entire season's capital investment and driving rural farming families into severe debt.

THE AQUACULTURE DISEASE CRISIS	
Global Annual Financial Losses	$6.0+ Billion USD per year
Smallholder Farm Representation	>85% of total global producers
Diagnostic Laboratory Lead Time	3 to 5 business days (too late)
Reactive Chemical/Antibiotic Cost	15% to 20% of total farm OpEx
Current Industrial Sensor Cost	$10,000 – $50,000 (Out of reach)

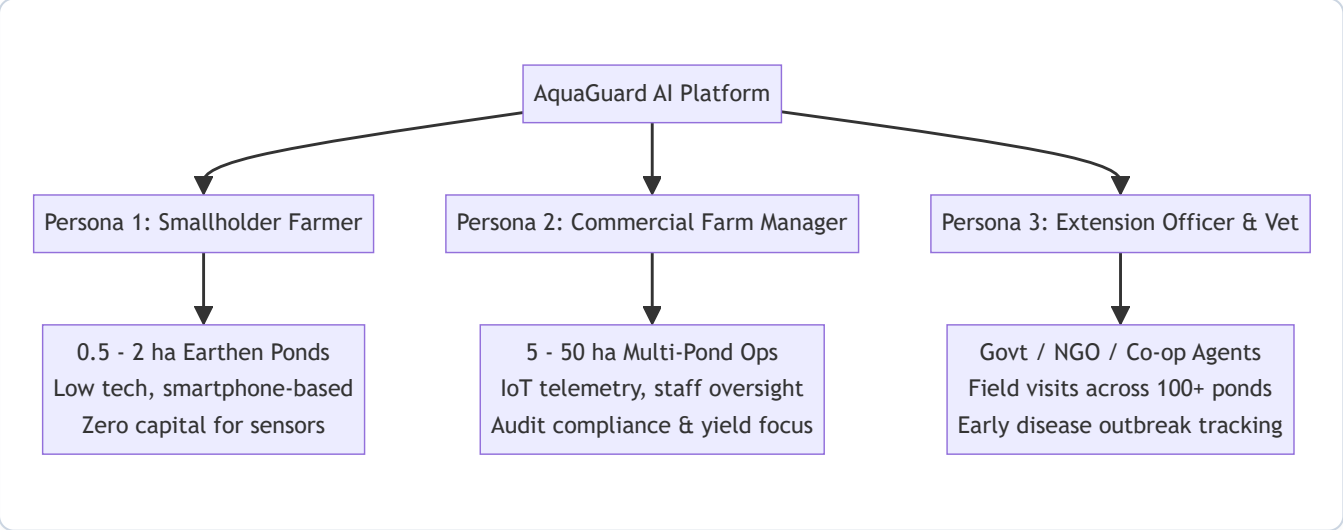
1.2 Market Sizing (TAM, SAM, SOM)

- **Total Addressable Market (TAM): $310 Billion USD** — The entire global aquaculture production market, spanning freshwater finfish, marine finfish, crustaceans, and mollusks across 130+ million metric tonnes of annual output.
- **Serviceable Addressable Market (SAM): $38.4 Billion USD** — The global freshwater pond aquaculture health management, water monitoring, feed diagnostics, and veterinary bio-security sector, heavily concentrated in developing nations (India, Bangladesh, Vietnam, Indonesia, Philippines, Nigeria, Egypt).
- **Serviceable Obtainable Market (SOM): $240 Million USD** — The addressable segment of small- to medium-scale commercial and semi-intensive freshwater pond farmers in South and Southeast Asia seeking smartphone-accessible, low-cost diagnostic software and early-warning intelligence over a 3-year horizon.

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2. Target Customer Personas & Market Segments

AquaGuard AI serves three distinct customer archetypes with specific operational realities:



2.1 Persona 1: "Ramesh" – Smallholder Fish Farmer (Primary Target)

- **Demographics:** Age 38, operates two 1-acre earthen ponds in Odisha, India; cultivates Indian Major Carps (Rohu, Catla) and Nile Tilapia.
- **Technology Profile:** Owns a low-cost Android smartphone (4G connectivity, intermittent in rural fields); uses WhatsApp and YouTube; limited English fluency (prefers local language or visual icons).

- **Workday Reality:** Inspects ponds at dawn and dusk; monitors feeding behavior from pond dykes; occasionally catches fish using a hand scoop net to check growth.
- **Core Frustrations:**
- Cannot see beneath the murky pond water; only notices disease when dead fish float to the surface.
- Cannot afford $1,000 electronic dissolved oxygen meters; relies on subjective water smell or seeing fish gasp at the surface.
- Local chemical dealer sells expensive antibiotics that often don't work or poison the pond.
- **Jobs to be Done (JTBD):** "*When I net a sluggish fish with spots or frayed fins, I want to take a quick photo with my phone and know immediately what disease it is, what water parameter went wrong, and exactly what affordable treatment (salt, lime, water exchange) to apply today so I don't lose my harvest.*"

2.2 Persona 2: "Vikram" – Commercial Pond Farm Manager

- **Demographics:** Age 45, manages a 25-hectare commercial semi-intensive farm with 12 distinct nursery and grow-out ponds in Andhra Pradesh, India.
- **Technology Profile:** Proficient with web dashboards, laptop and smartphone; utilizes digital accounting software; open to IoT sensors.
- **Core Frustrations:**
- Farm laborers fail to record water parameters accurately or miss early signs of fin erosion across distant ponds.
- Disease outbreaks in one nursery pond quickly spread farm-wide via shared equipment or inlet canals.
- Export processing facilities enforce strict zero-tolerance testing for antibiotic residues (chloramphenicol, nitrofurans); a single contaminated consignment causes complete shipment rejection.
- **Jobs to be Done (JTBD):** "*I want a single centralized dashboard to monitor all 12 ponds, track daily water quality trends, receive automated alerts when disease risk rises, and enforce non-antibiotic biosecurity protocols across my workforce.*"

2.3 Persona 3: "Dr. Ananya" – Fisheries Extension Officer / Cooperative Vet

- **Demographics:** Age 32, government field officer overseeing 150+ smallholder aquaculture ponds across a rural district.
- **Technology Profile:** Uses laptop and tablet; frequently travels between remote villages with spotty cellular coverage.
- **Core Frustrations:**
- Overburdened; cannot physically visit every pond before disease spreads across the river basin.
- Farmers call her after fish have already been dead for 48 hours, making post-mortem diagnosis difficult.
- Lacks rapid field diagnostic kits; sending tissue samples to the state university takes 5 days.
- **Jobs to be Done (JTBD):** "*I want a standardized mobile diagnostic screening tool that farmers can use in the field to submit structured visual and water reports, enabling me to triage regional outbreaks and recommend interventions remotely.*"

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3. Customer Pain Points & Unmet Market Needs

Pain Point ID	Real-World Problem	Current Market Workaround	Business & Operational Impact
PP-01: Optical Invisibility	Freshwater pond water is turbid (algae/silt); sub-surface symptoms are invisible until fish surface near death.	Naked-eye observation from pond embankments during feeding.	Catastrophic latency; 40–80% mortality before disease is recognized.
PP-02: Sensor Cost Barrier	High-precision commercial optical DO and multiparameter sondes cost $1,500–$5,000+ each.	Farmers use no sensors or buy unreliable $15 chemical titration drop kits that expire quickly.	Water crashes (nocturnal hypoxia, ammonia spikes) go completely undetected until mass suffocation occurs.
PP-03: Laboratory Vacuum	Diagnostic veterinary labs are located in urban university centers 50–200 km away from rural farms.	Farmers consult local feed/chemical salesmen who push high-margin proprietary chemicals.	Wrong treatments applied; 15–20% of operating expenditure wasted on ineffective chemicals.
PP-04: Antibiotic Dependency & AMR	Blind deployment of human and veterinary antibiotics (oxytetracycline, ciprofloxacin) as a panic response.	Farmers dump antibiotics directly into pond water without veterinary prescriptions.	Proliferation of drug-resistant pathogens; biofilter bacteria collapse; international export bans.
PP-05: Siloed Diagnostics	Existing tools only measure water OR classify images, never correlating the two.	Manual guesswork trying to connect a water test result with a skin lesion.	Misses the biological reality: poor water quality is the direct trigger that causes pathogens to attack.

4. Competitive Landscape & Positioning

4.1 Competitor Analysis Matrix

Solution	Target Market	Key Technologies	Pros	Cons & Market Gaps	Pricing
Aquabyte (Norway/US)	Industrial Atlantic Salmon (Marine Net Pens)	Submersible stereo HD cameras, deep CNNs, edge computing	Highly automated sea lice counting, biomass calculation	Marine cages only; completely ineffective in turbid freshwater ponds; extreme cost.	**Enterprise Subscription:** $2,000 – $5,000 / cage / month
XpertSea (Canada)	Commercial Shrimp Hatcheries & Ponds	Optical imaging hardware, mobile computer vision	Accurate shrimp counting, size distribution, harvest timing	Requires specialized physical hardware bucket; focused on shrimp, not finfish disease triage.	**Hardware + SaaS:** $3,000 – $7,000 setup
eFishery (Indonesia)	Freshwater Carp & Tilapia Farmers	Smart automated feeder, acoustic appetite sensors, IoT telemetry	Excellent feeding efficiency, established distribution network	Primarily an automated feeder; basic water logging; lacks computer vision disease diagnostics.	**Feeder Lease / Purchase:** $200 – $500 per unit
Manual Drop Test Kits (API, Hanna, Merck)	All Aquaculture	Chemical reagent titration / colorimetric test strips	Low upfront cost ($20–$50); no electricity required	Highly subjective color matching; tedious manual testing; zero automated alerts or advice.	**Consumable Cost:** $20 – $50 per kit (lasts 30 tests)
AquaGuard AI *(Our Solution)*	**Smallholder & Commercial Freshwater Pond Farms**	**MobileNetV3 CV + Random Forest Telemetry + Multimodal Risk Fusion Engine**	**Zero hardware requirement; smartphone camera ready; multimodal risk scoring; step-by-step non-antibiotic action plans.**	**Requires initial network connection for cloud inference; relies on user-submitted photos.**	**Freemium MVP ($0 for smallholders; Tiered SaaS for commercial farms)**

4.2 AquaGuard AI Competitive Differentiation ("The Moat")

1. **Multimodal Environmental & Pathological Risk Fusion:** We are the only platform that explicitly links physical water chemistry (DO, pH, $Temp$, NH_3) with visual dermatological symptoms through an interpretable biological synergy engine.
2. **Zero Capital Expenditure (No Expensive Hardware Required):** Smallholders can use their existing smartphones. Water telemetry can be entered via interactive UI sliders (using simple farm drop kits) OR streamed from low-cost IoT nodes.
3. **Prescriptive, Non-Chemical Agronomic Action Plans:** Rather than returning a useless raw label like "*82% Columnaris*", AquaGuard AI outputs immediate, step-by-step remedial protocols (e.g. "*Run paddlewheel aerator for 4 hours; apply 3 ppt non-iodized salt bath; withhold feeding today*").
4. **Offline-First & Mobile-Optimized:** Built with progressive caching and client-side image compression, functioning reliably even on weak rural 2G/3G networks.

5. Value Proposition & Business Model

5.1 Value Proposition by Stakeholder

- **For Smallholder Farmers:** *"Protect your entire season's fish harvest from mass mortality using just your smartphone, without buying expensive sensors or wasting money on useless antibiotics."*
- **For Commercial Farm Managers:** *"Gain real-time visibility across all your ponds, cut operational losses by up to 35%, ensure zero antibiotic residue rejections, and streamline staff workflows from a single dashboard."*
- **For Governments & Cooperatives:** *"Real-time regional biosurveillance map tracking disease vectors, preventing regional epidemics, and promoting sustainable, export-grade aquaculture biosecurity."*

5.2 Monetization & Business Model (Go-to-Market Strategy)

AQUAGUARD AI BUSINESS MODEL

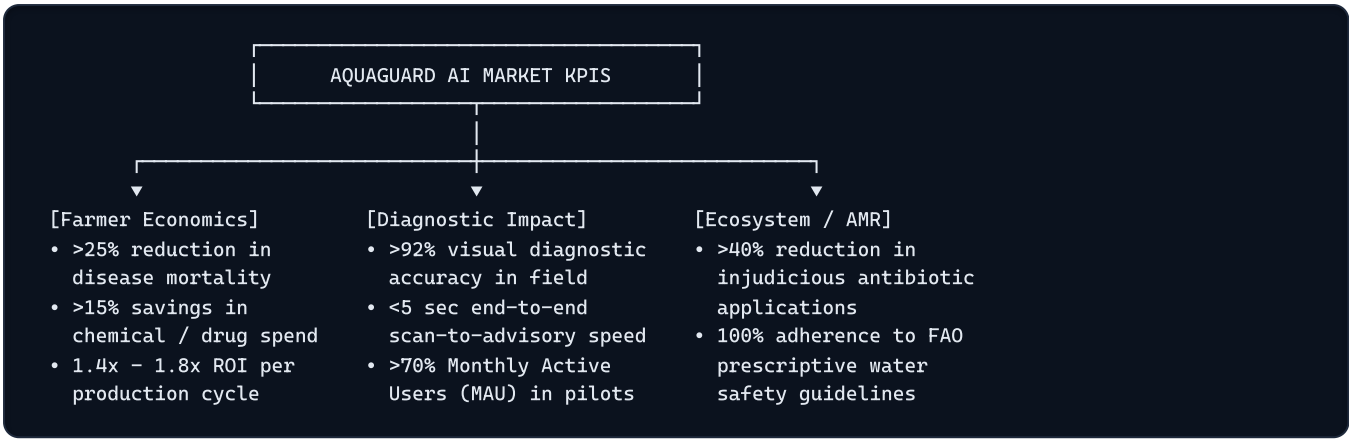
TIER	TARGET AUDIENCE	PRICING & FEATURES
1. Community (Freemium)	Smallholders (1–2 Ponds)	• FREE ($0 / month forever) • Up to 20 AI image scans/month • Manual water quality sliders • Standard biosecurity action plan
2. Pro Farm (Commercial)	Semi-Intensive & Commercial Farms (3–20 Ponds)	• $15 – $35 / farm / month • Unlimited AI scans across ponds • Automated IoT telemetry sync • Historical trend analytics • SMS / WhatsApp alert dispatch
3. Enterprise / Government (B2B & B2G)	Hatcheries, Feed Companies, State Fisheries Depts	• Custom Annual Contract • Regional disease epidemiology • White-label extension app • API access for farm sensor ERP

5.3 Distribution & GTM Partnerships

- **Phase 1 (Hackathon & Community Pilots):** Partner with local aquaculture cooperatives and university fisheries extension programs in Odisha and Andhra Pradesh (India) to onboard 50 smallholders for pilot testing.
- **Phase 2 (Feed & Fingerling Distributor Distribution):** Distribute AquaGuard AI access through rural fish feed distributors (e.g. CP Group, Godrej Agrovet, Cargill) who bundle the app as a value-add service to increase farmer harvest survival.
- **Phase 3 (Government Extension Integration):** Integrate with state fisheries department schemes (e.g., PMMSY - Pradhan Mantri Matsya Sampada Yojana in India) as a certified digital extension tool.

6. Market Success Metrics & Key Performance Indicators (KPIs)

To evaluate market viability and real-world impact, AquaGuard AI defines the following operational KPIs:



7. Regulatory, Environmental, and Biosecurity Compliance

1. **Antimicrobial Resistance (AMR) & One Health Alignment:** AquaGuard AI aligns directly with the WHO/FAO/WOAH "One Health" Global Action Plan by prioritizing non-chemical, water remediation, and biosecurity interventions over antibiotic treatments.
2. **Diagnostic Disclaimer & Veterinary Compliance:** AquaGuard AI is marketed strictly as a **"Risk Screening and Early Warning Decision Support Tool"**, not an official certification of animal health. Suspected high-mortality viral pathogens (TiLV, KHV) automatically trigger guidance directing farmers to certified government veterinary diagnostic laboratories.
3. **Data Privacy & Agronomic Security:** Farmer pond data, geolocation, and yield metrics are encrypted via Supabase PostgreSQL Row-Level Security (RLS) to ensure farmer confidentiality and proprietary harvest data protection.